Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/22/2020
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metal Oxides		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/21/2017	**End Date:**	05/20/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work (Ed., 5/18/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0059 ; NNX17AH73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research is implementing microgravity based experiments using the JAXA (Japanese Aerospace Exploration Agency) flight Electrostatic Levitation Furnace (ELF) to achieve precise measurements of the thermophysical properties of supercooled (also called undercooled) molten metal oxides. The work will include both ground based and flight experiments on liquids formed from aluminum, calcium, silicon, and rare earth element oxide mixtures. These liquids are precursors to high value-added glass materials that are used in photonics, lasers, optical communications, and imaging applications. The outcome of the proposed international research collaboration will be to: (i) Measure technologically important thermophysical properties of normal and supercooled liquid metal oxides of interest in connection with production of photonic and optical device materials, (ii) Develop thermophysical property datasets that complement existing atomic structure data on the materials under investigation. Use the resulting data to increase understanding of the nature of fragile liquids and the glass transition, and (iii) Disseminate the new data via publication, through NASA's Physical Sciences Informatics System and the Materials Genome Initiative to help accelerate the development of new products. We will investigate the behavior of liquid phase metal oxides as they transform to glass in quiescent, diffusion-controlled (i.e., without fluid motion) conditions. This work is needed to optimize photonic and optical device materials to obtain uniform properties and stable performance in applications. Overall the work will advance the broader goals of optimizing computational modeling of materials processes and developing superior materials by design. The flight ELF can achieve pristine molten and supercooled samples. Such samples can be studied using non-contact measurements of temperature, volume, surface tension, and viscosity. Spherical samples ca. 2 mm in diameter suitable for experiments in the flight ELF will be made by laser hearth melting. Prior ground-based work demonstrated that the sample materials can be: (i) levitated using electrostatic methods, (ii) heated using laser beams, and (iii) melted without significant evaporation. Proposed ground-based work will push the limits of these measurements and help to optimize the methods for analysis for the flight data. ED. NOTE 5/18/2020: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work relates to development, characterization, and optimization of functional glasses and glass ceramics. These materials are high value add products that are using in telecommunications, displays (TVs and phones), medical lasers, and security sensor equipment. The project work includes outreach to undergraduate and high school students.
Task Progress:	This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, laser and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. The project aims to measure the viscosity, density, and surface tension of a variety of molten oxides some of which form glass. Containerless conditions are essential in order to achieve deep supercooling without induced nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500ºC in containerless conditions. By working in microgravity it is expected that nearly quiescent molten samples can be obtained. Induced oscillations can be used to study the fluid properties of liquid materials. Analysis of images can reveal the oscillation amplitude and power spectrum from which the required properties are calculated. The materials of main interest are molten metal oxides many of which form fragile liquids that can be made into glass provided the melts can be deeply supercooled. In some cases there is already a considerable amount of data on the atomic structure of the melts. An important goal of this research is to correlate the structure and properties (in particular viscosity and density) to advance models of the processing of materials. These models are important in advancing basic science of liquids and glasses and in development of high value add applied materials. 1. Status: The project team includes Materials Development Inc. (MDI) scientists and Professors Robert Hyers University of Massachusetts, Amherst, Jonghyun Lee, Iowa State Univeristy, Dr. Shinji Kohara, National Institute for Materials Science, Japan, and Dr. Takehiko Ishikawa, Japanese Aerospace Exploration Agency, Japan. Professors Hyers and Lee are supported under a subcontract from MDI. The collaborators from Japan are not funded from this grant. The grant was awarded on 5/21/2017 and assigned a new contract number in November 2017. The period of performance was extended from four to five years and the budget was reduced. First year funding was at a level sufficient to enable scoping of the work and preliminary ground-based experiments in preparation for a more detailed experimental campaign during the subsequent years. Second year funding was also at a reduced level sufficient for planning and starting to prepare for the Science Concept Review (SCR). 2. Technical Activities: Activities during the second year included: (i) completion of the Science Requirements Document, Science Concept Review presentation, and the Science Concept Review panel, (ii) responding to panel comments with written responses, (iii) synthesis of final set of test samples that were sent to JAXA for ground-based testing prior to launch, and (iv) additional ground-based experiments to characterize flight samples. (i) MDI submitted the Science Requirements Document (SRD) to NASA on 1/29/2020. The document provided background and context for the research and details of the sample materials, experiment implementation and protocol, contingencies, and data analysis in the context of executing the experiments aboard the International Space Station (ISS). Soon after submission of the SRD, NASA scheduled a panel review. The SCR was held on 3/23/2020 via a virtual meeting. The SCR panel provided written recommendations in mid-April. (ii) The panel assessment and comments were complementary and the project was recommended for flight. The panel provided some specific recommendations for the investigator team to consider but did not make flight contingent on these. MDI provided a written response to the recommendations. We propose to run the first sample holder in air as proposed and evaluate the results, potentially running holder/s 2 and/or 3 in argon. Sample rotation with be evaluated by analyzing video images of samples prior to melting. Small asymmetries will provide a means to observe rotation in many cases. A lithium borate composition sample was added to the selection. The JAXA team provided information on cleaning the ELF. The SRD was updated and submitted to NASA for final approval and sign off. (iii) Two gram batches of each composition were made from high purity metal oxide powders. A set of 2 mm diameter test samples were sent to JAXA on 7/7/2020 for ground based testing. On 8/17/2020, JAXA confirmed that all

	compositions have passed the vibration test. Additional ground based tests will be performed including laser beam heating at 980 nm and effects of humidity on sample stability. (iv) Additional ground based tests are in progress to evaluate materials. These include synthesis of glass samples that will be measured using x-ray and neutron diffraction, chemical analysis of samples and evaluation of the critical cooling rate for samples that can form glasses. Results of these measurements will provide final detailed plans for the thermal cycles that will be used in the flight experiments. The atomic structure data will be used in interpretation of the results of the flight experiments and enable a more thorough understanding of how glass forming liquids behave when they are supercooled. 3. Plans: Plans for the next project period are as follows: (i) Continue experiments at NASA Marshall Space Flight Center (MSFC) to establish levitation conditions using the modified ESL, (ii) Test sample stability to humidity, vibration and long term storage, (iii) Continue development of the Science Concept Review documentation and plan for the SCR, (iv) Prepare publications based on the research. 4. Publications/presentations: Presentations and publications are at different stages of production. Abstracts for talks have been submitted to the 2020 ISS R&D Conference 9/22-24/2020 and the American Society for Gravitational and Space Research (ASGSR) Annual Meeting 11/5-6/2020. Both meetings will be held virtually. Two papers are in publication [Ed. note March 2021: papers have been added to Bibliography section below--Benmore et al., 2020; Benmore et al., 2021].
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Benmore C, Alderman O, Benmore S, Wilke S, Weber R. "Small and wide angle x-ray scattering studies of liquid-liquid phase separation in silicate melts." ACS Earth and Space Chemistry. 2020 Oct 15;4(10):1888-94. https://doi.org/10.1021/acsearthspacechem.0c00170 [Note reported in August 2020 as "accepted for publication as of August 2020."] , Oct-2020
Articles in Peer-reviewed Journals	Alderman OLG, Benmore CJ, Weber JKR. "Consequences of sp2 to sp3 boron isomerization in supercooled liquid borates." Appl Phys Lett. 2020 Sep 28;117(13):131901. https://doi.org/10.1063/5.0024457 , Sep-2020